THINKING FAST AND SLOW

Understand how your mind works.
Make better decisions.

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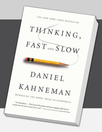

THE 2 SYSTEMS

Our brains work using 2 different systems.

System 1 (Fast Thinking)

- **Continuously** scans our environment.
- **Fast but error-prone**
- Works automatically & effortlessly via shortcuts, **impulses and intuition.**

System 2 (Slow Thinking)

- Used for specific problems, **only if necessary**
- Takes effort to analyze, reason, solve complex problems, **exercise self-control**
- **Slow but reliable**

System 1 is prone to biases & error; can be exploited by others to influence our responses & choices

Law of Least Effort

We're programmed to minimize effort. Since System 2 uses more effort & energy, System 1 runs the show by default.

System 2 is supposed to monitor & scrutinize System 1, but often doesn't when it's "lazy" or overloaded.

HEURISTICS (MENTAL SHORTCUTS)

System 1 works using Heuristics.
These generate biases and errors that affect our choices.

Key Heuristics

Associations & Priming
When exposed to an idea, we're "primed" to think about associated idea(s), memories & feelings.

Cognitive Ease
When we're at ease (things seem familiar, effortless, clear), we let our guard down.

Judgements
When making split-second judgements, we're better at broad averages and categorical comparisons than sum-like variables.

Stories & Causes
We construct stories to make sense of the world. We fill gaps with guesses & assumptions. The more coherent our story, the more confident we feel. But confidence ≠ accuracy.

Substitution
When a question is too complex, we unconsciously substitute it with a simpler question.

Emotions
When we emotionally prefer an outcome, we play up its benefits and play down its costs.

Heuristics cause Biases & Errors

Small Sample Sizes
We make decisions with inadequate data. Once a wrong conclusion is made, it triggers our associative mechanisms and multiplies the error.

Conjunction Fallacy
We perceive A+B to be more likely than A (when the reverse is true). We perceive an expensive item to be worth more on its own, than when a cheap item is added.

Availability
We overestimate an event's frequency if we're familiar with the idea (e.g. terrorism). Mass media multiplies this effect with vivid and repeated messages.

Causes Over Statistics
We tend to focus on causal data and neglect or ignore statistical data. We prefer stories with explanatory power, over mere data.

Regression to the Mean
We assign causal explanations to random fluctuations when such reasons don't exist.

Anchoring
Numbers can create "mental anchors" that bias our estimates & decisions.

Representativeness
We make judgments about a person, place, or thing based on how much it resembles something else.

Heuristics cause Overconfidence

Illusion of Understanding
We know the past less than we think we do. We'll never have all the details of an event to explain it accurately (the Narrative Fallacy) and once we change our views, we can't recall fully what we used to believe (the Hindsight Bias).

Illusion of Validity
We believe that skills produce predictable results – that's not always true. "Expert intuition" is only trustworthy if the environment is regular & predictable, and the expert has learned the regularities through prolonged practice.

Optimistic Bias
We tend to take risks based on best-case scenarios. We should validate our projections objectively against the "outside view".

Heuristics affect our Choices

Prospect Theory
We attach value to prospects of loss vs gain; We hate losing more than we like winning.

Keeping Score
Wins & losses involve emotions. Thus, we make irrational choices e.g. sell winners over losers, put good money after bad, and choose "safe" options for fear of regret.

Certainty & Possibility Effects
Generally, if we've a high chance for a desired outcome, we'd pay a premium to remove uncertainty (e.g. court settlement, insurance). We also may take irrational gambles for improbable outcomes (e.g. lottery tickets, terminal illness treatments).

Endowment Effect
An object that we own and use is more valuable to us.

Framing
Our frames shape our perceptions. We're more likely to choose options with "win" or "survive", than those with "lose" or "die.

Joint evaluations
Our decisions are different when done in isolation, vs when compared across scenarios.

OUR 2 SELVES

We have an "experiencing self" and a "remembering self". Our memories override our experiences, and we make decisions to create better memories (not better experiences).

Joint evaluations
We evaluate our life like a story via its "peaks" (key events and memorable moments) and how it "ends".

Joint evaluations
Do things that you enjoy, pay attention to your experiences, so you create more positive memories.

IN A NUTSHELL

Our minds use "heuristics" (mental shortcuts) to process inputs quickly. These are efficient but often flawed.

Know when you are prone to System 1 biases & errors => Slow down to activate System 2 when the stakes are high.